

## Literature Matrix

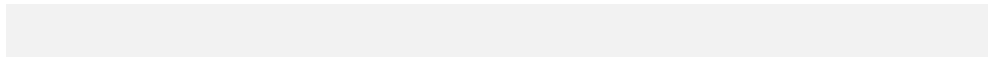
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| ID   | Year | Paper                                                                                                           | Category           |
|------|------|-----------------------------------------------------------------------------------------------------------------|--------------------|
| P001 | 2023 | The Rise and Potential of Large Language Model Based Agents: A Survey                                           | Survey             |
| P002 | 2023 | Generative Agents: Interactive Simulacra of Human Behavior                                                      | Multi-Agent        |
| P003 | 2022 | ReAct: Synergizing Reasoning and Acting in Language Models                                                      | Agent Architecture |
| P004 | 2023 | Toolformer: Language Models Can Teach Themselves to Use Tools                                                   | Agent Architecture |
| P005 | 2023 | VOYAGER: An Open-Ended Embodied Agent with Large Language Models                                                | Agent Architecture |
| P006 | 2023 | AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation                                        | Framework          |
| P007 | 2023 | CAMEL: Communicative Agents for “Mind” Exploration of Large Language Model Society                              | Multi-Agent        |
| P008 | 2023 | MetaGPT: Meta Programming for A Multi-Agent Collaborative Framework                                             | Framework          |
| P009 | 2025 | A Survey on Large Language Model based Autonomous Agents                                                        | Survey             |
| P010 | 2023 | MedAgents: Large Language Models as Collaborators for Zero-shot Medical Reasoning                               | Healthcare         |
| P011 | 2024 | Agent Hospital: A Simulacrum of Hospital with Evolvable Medical Agents                                          | Agent Architecture |
| P012 | 2024 | MedRAG: Enhancing Retrieval-augmented Generation with Knowledge Graph-Elicited Reasoning for Healthcare Copilot | Framework          |
| P013 | 2023 | Large Language Models Encode Clinical Knowledge                                                                 | Framework          |
| P014 | 2024 | A Survey of Large Language Models in Medicine: Progress, Application, and Challenge                             | Survey             |
| P015 | 2023 | Towards Expert-Level Medical Question Answering with Large Language Models                                      | Framework          |

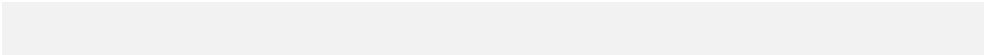
**Literature Matrix**

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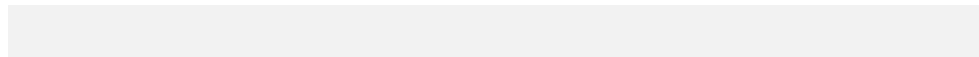
| ID   | Year | Paper                                                                                              | Category   |
|------|------|----------------------------------------------------------------------------------------------------|------------|
| P016 | 2023 | Towards Generalist Biomedical AI                                                                   | Framework  |
| P017 | 2023 | Towards Generalist Biomedical AI                                                                   | Framework  |
| P018 | 2018 | On Internal Categories and Crossed Objects in the Category of Monoids                              | Framework  |
| P019 | 2023 | Clinical Camel: An Open Expert-Level Medical Language Model with Dialogue-Based Knowledge Encoding | Healthcare |
| P020 | 2026 | AI Agents vs. Agentic AI: A Conceptual taxonomy, applications and challenges                       | Survey     |



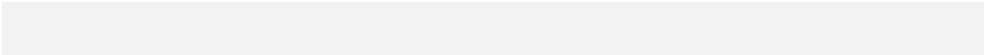
| Research Problem                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------|
| A lack of systematic understanding regarding the construction, application, and societal impact of LLM-based agents.                    |
| Creating believable proxies of human behavior that maintain long-term coherence in open-world environments.                             |
| LLMs struggle to combine reasoning traces with task-specific actions, limiting dynamic external interaction.                            |
| LLMs lack precise math, temporal awareness, and factual lookup abilities without heavy human annotation.                                |
| LLM agents lack lifelong learning capabilities for acquiring diverse skills in open-ended worlds.                                       |
| Scaling LLMs for complex tasks requiring the coordination of diverse agents, tools, and human inputs.                                   |
| Autonomous agent cooperation is hindered by role-flipping, conversation deviation, and reliance on heavy human prompting.               |
| Complex multi-agent task execution often suffers from logic inconsistencies and cascading hallucinations.                               |
| Existing agents lack human-like decision capabilities due to restricted training environments and isolated knowledge.                   |
| LLMs struggle with domain-specific terminologies and complex reasoning in the medical field without extensive fine-tuning.              |
| Enabling doctor agents to acquire medical expertise autonomously from practice without relying on manual data labeling.                 |
| Existing medical RAG models lack diagnostic specificity and accuracy, struggling to differentiate diseases with similar manifestations. |
| The lack of comprehensive benchmarks to thoroughly evaluate LLMs' clinical knowledge, reasoning capabilities, and patient safety.       |
| A scarcity of comprehensive reviews detailing the development, practical clinical deployment, and challenges of medical LLMs.           |
| Previous LLMs failed to match physician-level performance and struggled to safely answer complex, long-form clinical questions.         |



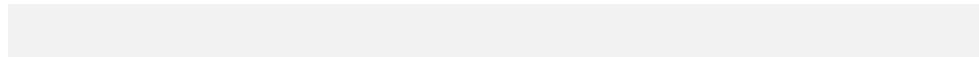
| Research Problem                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------|
| Most medical AI models are unimodal, single-task systems unable to flexibly integrate diverse multimodal patient data.              |
| Absence of comprehensive multimodal medical benchmarks restricts the development of unified, general-purpose biomedical AI systems. |
| The property describing internal categories for groups via semi-direct products fails in the category of monoids.                   |
| Proprietary medical LLMs raise privacy concerns, while existing open models underperform and have restricted context lengths.       |
| Literature conflates basic AI Agents with Agentic AI, causing conceptual ambiguity and misaligned system design.                    |



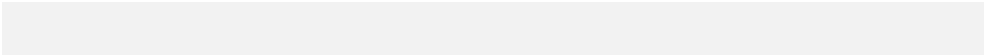
| Proposed Solution                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------|
| Proposes a comprehensive taxonomy and framework comprising brain, perception, and action modules for agents.                             |
| An architecture utilizing a memory stream, retrieval, planning, and reflection to simulate human behavior.                               |
| Interleaves reasoning traces and task-specific actions to induce, track, and update plans while gathering external information.          |
| A self-supervised method where LLMs autonomously learn to insert API calls to external tools.                                            |
| An agent with an automatic curriculum, an executable code skill library, and iterative prompting.                                        |
| A generalized framework employing customizable agents and a "conversation programming" paradigm to unify multi-agent workflows.          |
| A role-playing framework utilizing inception prompting to guide autonomous multi-agent conversations toward task completion.             |
| A meta-programming framework encoding Standardized Operating Procedures (SOPs) into role-based agent collaborations.                     |
| A comprehensive architectural framework detailing agent profiling, memory, planning, and action modules.                                 |
| A zero-shot multi-disciplinary collaboration framework simulating multi-round consultations among specialized LLM experts.               |
| Agent Hospital, a simulacrum where agents autonomously evolve by treating patient agents and reflecting on clinical outcomes.            |
| MedRAG, integrating a hierarchical diagnostic knowledge graph with RAG to elicit structured reasoning and proactive patient questioning. |
| Introduces the MultiMedQA benchmark and uses instruction prompt tuning to create Med-PaLM for safe medical QA.                           |
| Provides a systematic overview of medical LLM principles, compares task performance, and offers practical deployment guidelines.         |
| Med-PaLM 2, which leverages an improved base model, medical domain finetuning, and an ensemble refinement prompting strategy.            |



| Proposed Solution                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------|
| Med-PaLM M, a generalist multimodal generative model interpreting clinical language, imaging, and genomics with unified weights.             |
| Curates MultiMedBench to train and evaluate Med-PaLM M, unifying language, imaging, and genomics interpretation.                             |
| Introduces "crossed semi-bimodules" as a mathematical structure to construct and describe internal categories within monoids.                |
| Clinical Camel, an open LLM fine-tuned using Dialogue-Based Knowledge Encoding to convert texts into conversational data.                    |
| A structured taxonomy distinguishing single-entity tool-augmented agents from orchestrated, collaborative multi-agent Agentic AI ecosystems. |

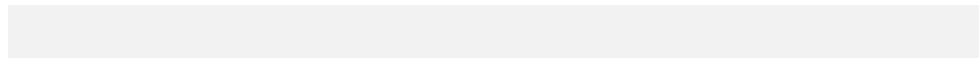


| Memory                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------|
| Yes – reviews short-term, episodic, semantic, and vector memories.                                                   |
| Yes – comprehensive natural language memory stream with recency, relevance, and importance retrieval.                |
| No explicit memory module; relies on prompt context.                                                                 |
| No                                                                                                                   |
| Yes – maintains a persistent skill library (vector database) of executable code.                                     |
| No explicit persistent memory; relies on conversational context.                                                     |
| Yes – maintains conversational history message sets iteratively during sessions.                                     |
| Yes – shared global message pool with a publish-subscribe mechanism for extracting relevant information.             |
| Yes – details short-term, long-term, unified, and hybrid memories with reading and reflection operations.            |
| No explicit persistent memory; relies on the consultation context and synthesized reports.                           |
| Yes – Maintains a medical case base for successes and an experience base for rules derived from failures.            |
| Yes – Uses FAISS-indexed EHR databases and a structured four-tier diagnostic knowledge graph for persistent context. |
| No explicit persistent memory architecture described; relies entirely on the model's parametric knowledge.           |
| Yes – Reviews systems integrating external databases and retrieval-augmented generation to mitigate knowledge decay. |
| No explicit persistent memory mechanism; relies on prompt context and model parameters.                              |

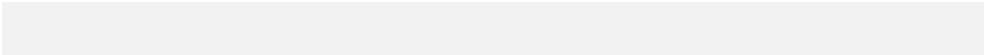


| Memory                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------|
| No explicit persistent memory mechanism; relies on the model's parametric knowledge and prompt context.           |
| No explicit memory architecture described; relies entirely on the multimodal sequence-to-sequence prompt context. |
| No                                                                                                                |
| No explicit persistent memory; relies entirely on the expanded 4096-token conversational context window.          |
| Yes – explores shared semantic, episodic, and vector-based persistent memory for distributed state management.    |

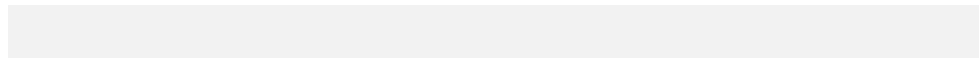




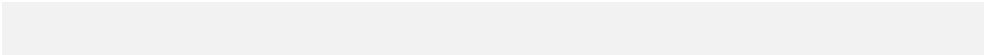
| Planning                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------|
| Yes – covers task decomposition, plan formulation, and plan reflection.                                                      |
| Yes – synthesizes memories into daily schedules and sequential action plans.                                                 |
| Yes – dynamically tracks and updates action plans via reasoning traces.                                                      |
| No explicit planning module.                                                                                                 |
| Yes – iterative prompting utilizing execution feedback and self-verification.                                                |
| Yes – supports task decomposition and hierarchical control for goal-directed workflows.                                      |
| Yes – an AI user agent acts as a task planner to interactively decompose execution steps.                                    |
| Yes – structured role-based workflow breaking down tasks into sequential SOP-driven subtasks.                                |
| Yes – summarizes single-path, multi-path, and feedback-driven task decomposition strategies.                                 |
| Yes – structured five-stage pipeline including expert gathering, analysis, summarization, consultation, and decision-making. |
| Yes – Simulates a complete closed-cycle workflow from triage and consultation to post-hospital follow-up.                    |
| Yes – Employs a proactive diagnostic questioning mechanism to sequentially gather missing patient information.               |
| No explicit planning or goal decomposition module implemented.                                                               |
| Yes – Discusses multi-agent collaborative planning for surgical tasks and dynamic motion planning for medical robotics.      |
| No explicit planning or sequential goal decomposition module.                                                                |



| Planning                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------|
| No explicit sequential planning or goal decomposition modules mentioned.                                                                  |
| No autonomous task decomposition or planning mechanisms implemented.                                                                      |
| No                                                                                                                                        |
| No explicit multi-step planning or goal decomposition modules.                                                                            |
| Yes – analyzes hierarchical goal decomposition, multi-step execution, and simulation-based planning governed by meta-agent orchestrators. |



| Reasoning                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------|
| Yes – reviews chain-of-thought and logical reasoning methods.                                                                      |
| Yes – recursive reflection to synthesize high-level inferences from observations.                                                  |
| Yes – uses chain-of-thought reasoning interleaved with environment actions.                                                        |
| Yes – self-supervised evaluation of when tool outputs help token prediction.                                                       |
| Yes – chain-of-thought reasoning for self-verification and code generation.                                                        |
| Yes – uses ReAct-style loops and implicit state inference for problem-solving.                                                     |
| Yes – instruction-following and Critic-in-the-Loop decision-making for complex task resolution.                                    |
| Yes – utilizes ReAct-style reasoning to navigate standard operating procedures.                                                    |
| Yes – reviews CoT, ToT, GoT, and other algorithmic reasoning methodologies.                                                        |
| Yes – enhanced via collaborative debates to reach unanimous consensus on clinical decisions.                                       |
| Yes – Retrieves past cases and experiences to reason through medical examinations, diagnosis, and treatment planning step-by-step. |
| Yes – KG-elicited reasoning synthesizes retrieved EHRs and diagnostic differences to accurately distinguish similar diseases.      |
| Yes – Elicits step-by-step clinical problem-solving using chain-of-thought and self-consistency prompting techniques.              |
| Yes – Reviews the use of CoT and ICL to simulate diagnostic thought processes and logic.                                           |
| Yes – Introduces ensemble refinement, where the model conditions on its own generated reasoning paths to answer.                   |

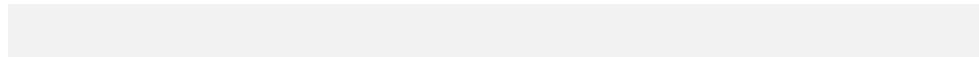


| Reasoning                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------|
| Yes – exhibits zero-shot multimodal chain-of-thought (CoT) reasoning to identify and describe medical findings step-by-step. |
| Yes – demonstrates zero-shot medical reasoning and language-enabled combinatorial generalization to novel clinical concepts. |
| No                                                                                                                           |
| Yes – trained to explicitly justify reasoning paths and explain why specific medical options are correct.                    |
| Yes – reviews causal modeling, reflexive self-critique, and tool-augmented reasoning across complex agentic workflows.       |

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| Tool Use                                                                                                       | Multi-Agent | RAG | Healthcare |
|----------------------------------------------------------------------------------------------------------------|-------------|-----|------------|
| Yes – discusses external APIs and physical embodiment tools.                                                   | Yes         | Yes | Yes        |
| No                                                                                                             | Yes         | Yes | No         |
| Yes – Wikipedia search and lookup APIs.                                                                        | No          | Yes | No         |
| Yes – calculator, Q&A, Wikipedia search, machine translation, calendar.                                        | No          | Yes | No         |
| Yes – uses JavaScript APIs for motor control and environmental action.                                         | No          | Yes | No         |
| Yes – Python execution, web search, Wolfram Alpha, and vector databases.                                       | Yes         | Yes | No         |
| Yes – HuggingFace agents for code execution and image generation.                                              | Yes         | No  | Yes        |
| Yes – Python code compilation, execution, and web search APIs.                                                 | Yes         | No  | No         |
| Yes – surveys usage of external APIs, databases, and specialized models.                                       | Yes         | Yes | Yes        |
| No                                                                                                             | Yes         | No  | Yes        |
| Medical knowledge bases, map editor (Tiled), and game framework (Phaser) utilized for building the simulacrum. | Yes         | Yes | Yes        |
| Yes – FAISS library for approximate nearest neighbor search over extensive Electronic Health Record databases. | No          | Yes | Yes        |
| No external tools or APIs utilized during model inference.                                                     | No          | No  | Yes        |
| Yes – Surveys the use of external knowledge bases, search engines, and robotic control APIs.                   | Yes         | Yes | Yes        |
| No external tool use mentioned in the methodology.                                                             | No          | No  | Yes        |

| Tool Use                                                                                                | Multi-Agent | RAG | Healthcare |
|---------------------------------------------------------------------------------------------------------|-------------|-----|------------|
| No external tools or APIs utilized during model inference.                                              | No          | No  | Yes        |
| No external tools or APIs mentioned in the methodology.                                                 | No          | No  | Yes        |
| No external tools utilized.                                                                             | No          | No  | No         |
| No external tools or APIs utilized during inference.                                                    | No          | No  | Yes        |
| Yes – details integration of web search, function calling, and APIs for real-time external interaction. | Yes         | Yes | Yes        |

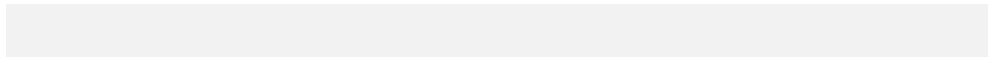


| Trustworthy AI                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------|
| Yes – explores adversarial robustness, trustworthiness, and ethical governance.                                                 |
| Yes – addresses ethical risks of parasocial relationships and deepfakes.                                                        |
| Yes – improves human interpretability and reduces hallucinations.                                                               |
| Yes – reduces factual hallucinations using external knowledge tools.                                                            |
| Yes – implements self-verification to ensure safe and correct task completion.                                                  |
| Yes – employs specialized safeguard agents to verify code safety and prevent harmful outputs.                                   |
| Yes – aligns autonomous collaboration with human intentions using AI alignment and critic loops.                                |
| Yes – implements executable feedback and cross-role code reviews to reduce hallucinations and bugs.                             |
| Yes – reviews challenges in AI alignment, prompt robustness, and hallucination mitigation.                                      |
| Yes – collaborative synthesis mitigates CoT-induced hallucinations and corrects initial diagnostic inconsistencies.             |
| Yes – Reflects on failed diagnoses to generate corrective rules, improving future safety and accuracy.                          |
| Yes – Grounds responses in verifiable EHR data and knowledge graphs to explicitly reduce misdiagnosis risks.                    |
| Yes – Rigorous human evaluation framework assessing factuality, potential for harm, bias, and scientific consensus alignment.   |
| Yes – Comprehensively addresses challenges like hallucinations, ethical concerns, bias, and the need for regulatory frameworks. |
| Yes – Evaluated against challenging adversarial datasets specifically designed to probe demographic bias and potential harm.    |

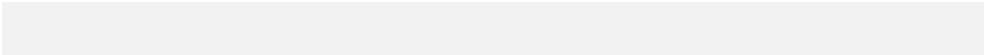


| Trustworthy AI                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------|
| Yes – includes blinded radiologist evaluations to assess clinical efficacy, safety implications, and error rates.             |
| Yes – performs independent human evaluations to rigorously annotate clinical errors, omissions, and hallucinations.           |
| No                                                                                                                            |
| Yes – emphasizes proactive bias detection, safety evaluations, and open transparency to ensure ethical clinical deployment.   |
| Yes – addresses critical needs for ethical governance, accountability, bias mitigation, and verifiable multi-agent alignment. |

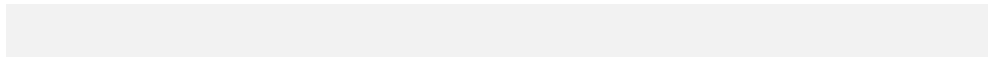




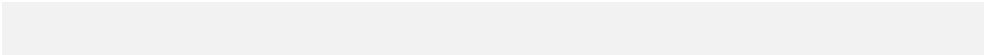
| Novelty / Key Innovation                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------|
| Comprehensive synthesis of agent capabilities from isolated entities to orchestrated multi-agent societies.                             |
| Fusing LLMs with a dynamic memory stream for long-term multi-agent social simulations.                                                  |
| Synergizing reasoning and acting in a single interleaved LLM prompt loop.                                                               |
| Self-supervised learning of tool use via API calls without massive human-annotated datasets.                                            |
| Using an executable code skill library to achieve gradient-free lifelong learning.                                                      |
| Flexible conversation patterns enabling dynamic topologies for humans, tools, and agents to interact seamlessly.                        |
| Inception prompting and role-playing for scalable task-solving with minimal human intervention.                                         |
| Integrating human-like SOPs and an assembly line paradigm into LLM-based multi-agent systems.                                           |
| Provides a systematic, holistic taxonomy of LLM-based agent construction, capability acquisition, and applications.                     |
| Zero-shot, training-free extraction of intrinsic medical knowledge via multi-disciplinary expert role-playing and consensus protocols.  |
| Introduces the Simulacrum-based Evolutionary Agent Learning (SEAL) paradigm for gradient-free autonomous medical expertise acquisition. |
| Combines RAG with knowledge graph-elicited reasoning and proactive questioning to highly specify diagnostic decisions.                  |
| Creates a comprehensive, human-centric evaluation framework for medical LLMs and introduces instruction prompt tuning.                  |
| Delivers a holistic taxonomy of medical LLMs, practical deployment guidelines, and extensive performance comparisons.                   |
| The ensemble refinement prompting strategy and achieving state-of-the-art, expert-level performance on the USMLE benchmark.             |



| Novelty / Key Innovation                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------|
| First generalist biomedical AI system handling a diverse range of multimodal tasks with a single model.                                |
| Curates MultiMedBench (14 tasks) to facilitate training single-weight generalist models across disparate medical domains.              |
| Formally defines crossed semi-bimodules and relates them mathematically to crossed semi-modules and internal categories.               |
| Dialogue-Based Knowledge Encoding (DBKE) synthesizes complex medical literature into conversational training data for soft alignment.  |
| First formal taxonomy delineating the architectural evolution from prompt-driven generation to decentralized Agentic AI orchestration. |



| Evaluation                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------|
| Literature review and comparative taxonomy formulation without empirical benchmarking.                                            |
| Evaluated via human-rated interviews and an end-to-end 25-agent sandbox simulation.                                               |
| Evaluated on HotpotQA, Fever, ALFWorld, and WebShop against standard baselines.                                                   |
| Zero-shot evaluation on math, QA, and temporal datasets showing significant improvements.                                         |
| Evaluated in Minecraft on item discovery and tech tree mastery against baselines.                                                 |
| Measured success rates on MATH, ALFWorld, and OptiGuide benchmarks against single-agent baselines.                                |
| Human and GPT-4 evaluations comparing multi-agent solutions to single-shot baselines across generated datasets.                   |
| Tested on SoftwareDev and HumanEval benchmarks via executability scores and human revision costs.                                 |
| Compares subjective and objective evaluation protocols, including human annotation, metrics, and simulations.                     |
| Evaluated zero-shot performance on MedQA, MedMCQA, PubMedQA, and MMLU subtasks against baseline LLM prompting.                    |
| Evaluated on MedQA (USMLE) and virtual tasks for examination, diagnosis, and treatment against state-of-the-art baselines.        |
| Assessed quantitatively on DDXPlus and a chronic pain dataset, alongside subjective physician evaluation of generated reports.    |
| Benchmarked on MultiMedQA and evaluated by expert clinicians for scientific consensus, reading comprehension, and potential harm. |
| Literature-based comparative analysis of LLM performance against lightweight models and human experts across various tasks.       |
| Evaluated on MultiMedQA using pairwise comparative ranking by physicians to assess clinical utility and safety.                   |

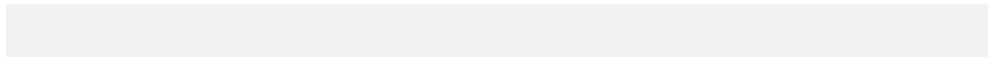


| Evaluation                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Quantitative metrics on 14 tasks (MultiMedBench) and qualitative radiologist ranking of generated chest X-ray reports.                        |
| Automated benchmarking against specialized SOTA models and independent human evaluation of omission and error rates.                          |
| Evaluated solely through abstract mathematical proofs verifying structural homomorphisms, identities, and category equivalences.              |
| Evaluated on USMLE, MedQA, and MedPubQA in zero- and five-shot settings against GPT-3.5 and GPT-4.                                            |
| Qualitative literature synthesis, architectural capability comparisons, and conceptual mapping across diverse real-world application domains. |

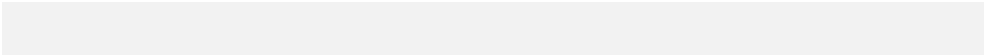
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| Research Gap                                                                                                                    | Relevance |
|---------------------------------------------------------------------------------------------------------------------------------|-----------|
| Identifies gaps in causal reasoning, formal safety verification, and scalable multi-agent coordination.                         | 9         |
| High computational token costs and occasional memory hallucination or embellishment.                                            | 7         |
| Potential risks of taking harmful actions in real-world environments without safeguards.                                        | 8         |
| Inability to use tools interactively or chain multiple tool outputs together.                                                   | 8         |
| High API costs and difficulties processing pure spatial reasoning without visual inputs.                                        | 8         |
| Determining optimal, cost-effective multi-agent workflows and topologies for varying real-world complexities.                   | 9         |
| Overcoming conversation loops and safely aligning complex, open-ended autonomous cooperation with human intentions.             | 8         |
| Limited capacity for cross-project continuous learning and addressing diverse, complex real-world application requirements.     | 9         |
| Identifies unresolved challenges in hallucination, knowledge boundaries, and aligning agents with diverse human values.         | 9         |
| Requires continuous updating of parameterized knowledge and adaptation for low-resource medical languages.                      | 10        |
| The base models are frozen, limited to high-level treatments, and lack cross-departmental doctor consultations.                 | 10        |
| Requires integration of multimodal data (MRI, ECG) and real-time speech recognition during live consultations.                  | 10        |
| Needs to ground responses in authoritative sources, communicate uncertainty, and ensure equity across diverse demographics.     | 9         |
| Highlights the critical need to integrate multimodal time-series data (ECG) and develop comprehensive clinical benchmarks.      | 9         |
| Explicitly calls for exploring inter-physician variation and expanding adversarial datasets for broader health equity coverage. | 9         |

| Research Gap                                                                                                                         | Relevance |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Needs large-scale multimodal datasets to jointly scale encoders and rigorous validation in real-world clinical applications.         | 10        |
| Exploring the safety, equity, and performance impacts of scaling modality-specific encoders alongside language models.               | 10        |
| Application to rank two commutative algebras and Galois algebras requires further mathematical exploration.                          | 1         |
| Needs multi-modal capabilities, efficient knowledge updating via retrieval, and rigorous human evaluation across clinical scenarios. | 8         |
| Lack of standardized communication protocols, unresolved causal foundations, and the need for formal safety verification.            | 10        |



| Contribution to Thesis                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------|
| Provides the foundational theoretical taxonomy to structure the proposed clinical agent framework and its capabilities.              |
| Offers an architecture for maintaining long-term patient context and multi-agent clinical coordination via shared memory.            |
| Provides a foundational reasoning-acting loop allowing healthcare agents to iteratively plan and safely interact with clinical data. |
| Shows how clinical models could self-teach to query EHRs or medical calculators autonomously.                                        |
| The persistent skill library concept can inspire dynamic, lifelong updating of automated clinical protocols.                         |
| Provides flexible multi-agent topologies and conversation programming to orchestrate human-AI clinical workflows.                    |
| Demonstrates inception prompting to strictly maintain specialized clinical roles during autonomous patient data analysis.            |
| Demonstrates how Standardized Operating Procedures (SOPs) can ensure strict clinical guideline adherence in agentic frameworks.      |
| Offers the foundational architectural taxonomy required to logically structure the intelligent patient monitoring framework.         |
| Directly supplies the multi-agent clinical consultation and consensus mechanism central to the proposed decision support system.     |
| Provides a scalable simulation framework for medical agents to continuously learn and evolve from patient interactions.              |
| Demonstrates how structured knowledge graphs and RAG can improve diagnostic specificity and enable proactive patient monitoring.     |
| Establishes the rigorous human-centric evaluation metrics essential for validating the safety of clinical decision support agents.   |
| Offers a complete taxonomy of capabilities, challenges, and architectural guidelines to structure a clinical agent framework.        |
| Introduces ensemble refinement for improved diagnostic reasoning and highlights the necessity of adversarial safety testing.         |



| Contribution to Thesis                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------|
| Provides a foundational multimodal architecture capable of seamlessly integrating diverse patient data streams for clinical monitoring.   |
| Demonstrates the feasibility of using a single generalist model to process complex multimodal clinical inputs.                            |
| Purely focused on abstract algebra and category theory; offers no contribution to clinical decision support.                              |
| Provides an open-source, high-performing conversational engine suitable for privacy-compliant intelligent patient monitoring deployments. |
| Supplies the definitive multi-agent architectural taxonomy and coordination framework required to structure the decision support system.  |